Reconstruction and linearization in *wh*-movement of DPs[‡]

Abstract

The goal of this paper is to propose an account of reconstruction in *wh*-movement without moving the entire *wh*-phrase and then applying scattered deletion at LF. The conceptual motivation for such an approach is that, in *wh*-movement, the *wh*-determiner has to be interpreted only in the higher copy, for the purposes of question semantics, and, the restrictor of this *wh*-determiner must be interpreted in the lower copy (for interpretive reasons, such as binding). There is a way to build such structures in the narrow syntax directly and then solve the problem of pronouncing the restrictor of the *wh*-determiner in the higher position by writing a linearization algorithm. I argue that this linearization algorithm, although relatively complex, puts us in a better position than the scattered deletion approach because it can explain otherwise unexplainable reconstruction patterns.

Word count: 10839

Keywords: Neglect, Late Merge, reconstruction, multidominance, linearization, *wh*-movement

1 Introduction

The goal of this paper is to address some complex patterns in cases of so-called *reconstruction*. For instance, under a mainstream understanding of the *wh*-movement in (1a), *picture of herself* needs to be c-commanded by *every girl*, so *herself* can be bound. That is, for the sake of interpretation, the structure for (1a) must be close to (1b). However, there is a distinct sense in which the entire *wh*-phrase moves. The traditional response to this is to move the whole *wh*-phrase in narrow syntax and apply scattered deletion at LF. This, in a sense, seems an easy enough thing to do because the linear order that is heard in (1a) can be accounted for by always deleting the lower copy at PF. I want to persuade the reader to explore the opposite route: generate the structure in (1b) in narrow syntax directly. The trade-off is to write rules of linearization that are systematic although somewhat more complex, but breaking the tie in favor of semantics will remove other explanatory hurdles that arise from the deletion operations.

‡. Acknowledgements to be added.

(1) a. Which picture of herself$_1$ did every girl$_1$ find?

b. [which] did every girl$_1$ find [picture of herself$_1$]

I will begin by introducing some assumptions. For instance, in (2), *which picture of herself* is said to undergo *wh*-movement from the object position of the preposition *on* to SpecCP. With Heim (2019), I will assume that the LF for this sentence is (2c); that is, only *which* has to be LF-interpreted upstairs for reasons of question semantics, and *picture of herself* needs to be LF-interpreted downstairs, so that *every girl* can bind *herself* and satisfy Condition A of the Binding Theory.[1] However, because the entire *wh*-phrase is pronounced in its higher SpecCP position, the prominent tendency in how to represent this displacement in the syntax has been to outright assume that what moves is what is heard to be moving, that is, the entire *wh*-phrase. But, assuming the Copy Theory of movement for now, this simplistic approach usually has a hard time predicting and constraining which part of which copy of a moving phrase is interpreted at which interface. More specifically, what predicts that the lower copy is to be deleted at PF, and, the specific pattern of scattered deletion shown in (2c) has to take place at LF?

(2) Which picture of herself$_1$ did every girl$_1$ find?

a. ***Syntax:***
[which picture of herself$_1$] did every girl$_1$ find [which picture of herself$_1$]

b. ***PF:***
[which picture of herself$_1$] did every girl$_1$ find ~~[which picture of herself$_1$]~~

c. ***LF:***
[which ~~picture of herself$_1$~~] did every girl$_1$ find [~~which~~ picture of herself$_1$]

The easy way out of this is the standard approach, which is the following. At PF, it is the lower copy that is consistently chosen for deletion. At LF, on the other hand, various deletion patterns arise, but only those that yield semantically interpretable, that is, composable, structures are the ones that succeed. That is, there is no other option at PF than the one shown in (2b). But, for LF, various other deletion patterns are conceivable, but they would lead to some kind of semantic crash, most importantly, type mismatches. For instance, assuming *which* is an existential quantifier (in which I follow Heim 2019 among many others), if it is LF-interpreted in the lower copy, then that would yield an $\langle\langle e, t\rangle, t\rangle$-type object for *find* to combine with, and, assuming *find* is an $\langle e, \langle e, t\rangle\rangle$-type predicate, semantic composition will crash.[2] When all such deletion patterns are explored, it turns out that only the one shown in (2b) survives.

Similar reasoning can be offered for the opposite kind of phenomenon observed in *wh*-movement, that is, antireconstruction, in cases like (3b). (3a) is one of the many cases that

1. By *LF-interpreted*, I will mean "interpreted at/by LF" for the purposes of semantic composition. Similarly, by *PF-interpreted*, I will mean "interpreted at/by PF", for the purposes of linearization.

2. This is identical to the rationale for Quantifier Raising given in Heim & Kratzer (1998).

have long been known to demonstrate that the NP restrictor of an $\overline{\text{A}}$-moving DP is reconstructed, leading to the violation of Condition C, which causes (3a) to be ungrammatical (see van Riemsdijk & Williams 1981, Freidin 1986, Lebeaux 1988, Fox 1999, among others). However, it has also been observed that the entirety of this NP restrictor isn't subject to reconstruction. For instance, in (3b), the R-expression *Alma* in the CP *that Alma wrote* doesn't incur a Condition C violation, which means that, for the purposes of LF-interpretation, the structure of the sentence is such that *Alma* is not c-commanded by *she*.[3]

(3) a. *Which aspect of Alma$_1$ does she$_1$ despise?

b. Which **paper** that Alma$_1$ wrote did she$_1$ later publish?

There is a variety of explanations that have been proposed for this. The traditional explanation, championed in Lebeaux (1988), and represented in Lebeaux (1990, 2000, 2009), Chomsky (1993), Fox (1999, 2002, 2017), Fox & Nissenbaum (1999), Overfelt (2015), *inter alia*, is that there is Late Merge (henceforth, "LM"), that countercyclic Merge, of the relative clause (henceforth, "RC") to the NP of the moved DP *which paper*, as shown in (4). That is, *which paper* is base-generated as the object of *publish* and undergoes *wh*-movement. It is only after this that the RC enters the modification structure with *paper*. Because of this, there is no occurrence of *Alma* anywhere in the c-command domain of *she*, and Condition C is thereby satisfied.

(4) Which [paper [$_{\text{RC}}$ that Alma$_1$ wrote]]
did she$_1$ later publish ⟨which paper⟩?

However, various problems — mostly concerning overgeneration — arise if countercyclic Merge is allowed as an operation in the system of grammar. This has been discussed in Sportiche (2018); so I will not go over it here and take it for granted that LM is not a licit operation. On the face of this, then, the traditional LM account of (3b) needs an alternative. Sportiche (2016) offers an alternative account in terms of a more radical scattered deletion approach than the kind shown in (2) (which he calls *Neglect*). This involves base-generating the entire *wh*-phrase *which paper that Alma wrote* in the object position of *publish*, *wh*-moving it to SpecCP in the narrow syntax, and having the scattered deletion pattern in (5). The crucial part is that the RC is both PF-interpreted and LF-interpreted only in its higher position inside SpecCP, which allows *Alma* to never be LF-interpreted in the c-command domain of *she*.

3. For recent skepticism about these reconstruction facts, see Adger, Drummond & van Urk (2017) and Bruening & Al Khalaf (2019). For counterarguments against this body of work, see Stockwell, Meltzer-Asscher & Sportiche (2021, 2022). This boils down to whether there exists an argument-adjunct asymmetry in antireconstruction. This is not directly relevant to the discussion in this chapter, as the following sections will make precise; so, for now, I would set aside the variations in judgements that exist on this issue.

(5) a. ***PF:***
Which [paper [$_{\text{RC}}$ that Alma$_1$ wrote]]
did she$_1$ later publish ~~which [paper [$_{\text{RC}}$ that Alma$_1$ wrote]]~~?

b. ***LF:***
Which [paper [$_{\text{RC}}$ that Alma$_1$ wrote]]
did she$_1$ later publish ~~which~~ [paper ~~[$_{\text{RC}}$ that Alma$_1$ wrote]~~]?

Arguably, there is a certain systematicity in this scattered deletion pattern, just as the in (2): one could say that, at PF, the entire higher copy is consistently chosen for the purposes of linearization, but, at LF, depending on which part of which copy is deleted, different coreference possibilities arise and no separate constraint needs to be imposed on scattered deletion. For instance, if the entire NP restrictor including the RC were LF-interpreted in the lower copy, then a disjoint reference effect would arise. But nothing goes wrong when the RC is LF-interpreted only in the higher copy, which creates the additional possibility of coreference between *Alma* and *she*. That is, there is no randomness; the space of scattered deletion pattern is fully open, and the resulting structures are allowed or ruled out for independent reasons.

But I want to argue that this systematicity is only superficial and, in fact, any such scattered deletion approach is conceptually lacking. Notice that there is no principled reason for why any of the deletion operations discussed above *must* (and not just *can*) take place. The scattered deletion approach makes sense only to the extent that the possibility of deleting parts of copies at LF at all makes sense, for which there is no clear justification. This is because of a fundamental asymmetry between PF and LF considerations. It is understandable why PF deletion is a reasonable operation to posit; otherwise, every copy of every moving element must be pronounced and that will lead to way too unwieldy strings. But such a consideration simply does not apply at LF. That is, nothing goes wrong if *picture of herself* is not deleted in the higher copy, as shown in (2c). It is of the right type ($\langle e, t\rangle$) for *which* (usually assumed to be a generalized existential quantification determiner; type $\langle\langle e, t\rangle, \langle\langle e, t\rangle, t\rangle\rangle$) to combine with. In fact, deletion makes it unclear how *which* should combine with the rest of the structure (more on this later, once I have been more explicit about the semantics of *which*).

In the same way, nothing goes wrong for (5b) if the RC *that Alma wrote* is interpreted both in the higher position and the lower position, except for the disjoint reference effect between *Alma* and *she*. But this disjoint reference effect is not an anomaly or a fatal property of the sentence: it allows a coherent meaning to arise, in which these two DPs must refer to different individuals. That is, under a more reasonable view of scattered deletion (where its absence would have to be fatal in order for it to apply), the prediction should be a disjoint reference effect between *Alma* and *she*. Therefore, I take this to be a fundamental gap in the understanding of scattered deletion approaches. My goal in this paper is to find an alternative to this.

One aspect that is shared between the scattered deletion approach in (2) and that in (5) is that, in *wh*-movement, the entire *wh*-phrase moves as a unit. My goal in this paper is to argue that the very first step that needs to be taken in analyzing how *wh*-movement should be formalized is to go in the opposite direction to the one that is presupposed in the representations in (2). (2) presupposes that, in modeling *wh*-movement, what should be described as the moving element is what linearization makes it seem is moving, that is, the entire *wh*-phrase. This has been the general tendency perhaps because the impression made by pronunciation is too hard to ignore during analysis. That is, what linearization makes apparent is what is taken to be cue for what the theory might be. My proposal is to do just the opposite: to take the semantics as the cue for it. In other words, I will propose for the narrow syntax itself to a structure close to (6) for (2), and then relegate to PF the job of pronouncing *picture of herself* to the left.

(6) [which] did every girl$_1$ find [picture of herself$_1$]

In order to produce such structures and, at the same time, make it contain enough information for the PF to be able to linearize it, some structural link must be established between the content of SpecCP and the object of *find*. Under standard accounts, this link is the movement link itself. But, since my goal is to not move the whole DP, I will propose that this link is established by a node $\mathcal{R}$ (a domain restriction variable; more on this below) that is simultaneously sister to *which* and *picture of herself*, without the two being sisters of each other, as shown in (7). Differently put, this amounts to removing transitive closure from the notion of sisterhood.[4]

(7) [which $\mathcal{R}^1$] did every girl$_1$ find [THE $\mathcal{R}^1$ picture of herself$_1$]

I will implement this within a system of multidominance, under which movement will be understood as Remerge.[5] There are many aspects of these LFs that need to be explained, which I will do in the course of the paper. However, in the context of the discussion so far, it is clear why (7) yields identical effects as (2c). What remains to be explained is how PF yields the attested linearization. For that, I will propose an independently necessary linearization algorithm for moved DPs. This algorithm will apply to the LF in (7), producing the string in (1a).

This is the gist of the paper. Besides the problem of why scattered deletion is allowed at LF at all — which I discussed after introducing (5) — there are further reasons to think why the approach I am advocating here is superior to a more mainstream scattered deletion approach. Before I get to that, I will first elaborate on the minimal assumptions and details necessary for each part of this analysis to work and propose the linearization algorithm itself, and then, I will devote the rest of the paper to presenting what I believe is further empirical motivation for an approach like the one I am proposing here. These motivations involve what

4. I cosuperscript two things to indicate that, in the syntax, they are one and the same object in two different positions.

5. The system I will propose here are very close, if not identical, to the ones developed in Fox (2017), Fox & Johnson (2016), Johnson (2009, 2012, 2018). Interested readers might find this literature instructive to put my system into perspective, but here, I will not devote any space to the comparison of these systems.

seems to be complex but systematic patterns of scattered deletion that no current theory of such deletion operations can predict/explain. I will, therefore, not present these motivating data in the introduction. But to briefly preview the results, I will present some crucial data, which will serve to argue that the linearization algorithm argued for in this paper, although slightly more complex than what we would expect any linearization algorithm to be, is systematic and comes with explanatory power. The crux of the proposal will be that, because of the nature of the algorithm, there are strings which it can never generate, that is, strings which it can constructively eliminate.

I will now turn directly to laying out my approach. I will center this around the question of how to generate structures like (7). Once that is done and we have the different pieces of the framework, it will become easier to appreciate the motivating data and the explanatory power of the approach advocated here.

2 Alternative for moving the entire DP

2.1 In QR in general

To repeat, in *wh*-movement, what is usually analyzed as the moving element is the entire *wh*-movement. The alternative to this that I will propose here is to move the DP, yes, but move it by remerging nodes in it with each other all over again, thereby making them separable from each other, and then merge the new node created by these Remerge operations in the landing site. This will result in a single syntactic node having multiple mothers, *i.e.*, multidominance. Before I get to showing precisely how I envisage this to happen, I will first introduce some assumptions about DP movement and its semantics. Let us begin with (8).

(8) Mary saw every girl.

In order to encode the relevant semantic information for variable binding in the syntax of DP movement, I will assume, with Heim & Kratzer (1998), that the LF of (8) needs to at least vaguely correspond to the one schematized in (9), on some relevant level of abstraction.[6]

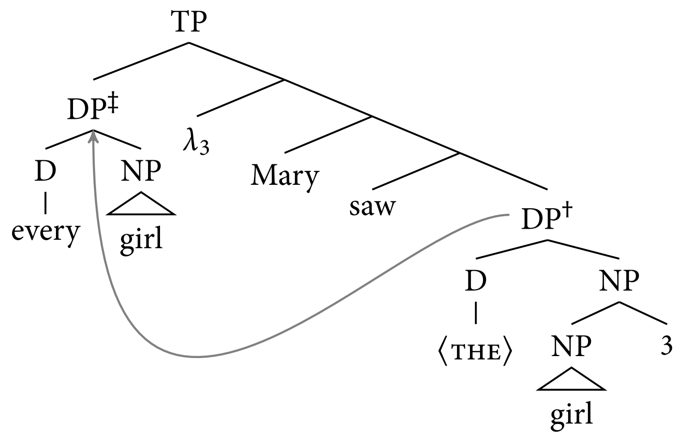
(9) [every girl] λ_3 [Mary saw t_3].

The aspect of (8) that (9) seeks to encode is the usual type-based logic of Heim & Kratzer (1998): *every girl* is of type $\langle\langle e, t\rangle, t\rangle$. Therefore, it cannot combine with *saw*, which is of type $\langle e, \langle e, t\rangle\rangle$. To resolve this type-mismatch, *every girl* needs to move from its base position, leaving an *e*-type trace. This *e*-type trace can now compose with *saw*. To make this story compatible with the Copy Theory of movement, it is usual to say that *every girl* is copied from the lower position to the higher position, and, through the separate step of Trace Conversion

6. Considerations of Scope Economy (Fox 1995, 1998, 2000) are ignored here for the sake of simplicity and demonstration.

(Fox 2002), the lower copy — which is still of type $\langle\langle e, t\rangle, t\rangle$ — is converted into an e-type trace. This is shown in (10), with Trace Conversion given in (11).

(10)



(11) **Trace Conversion**

a. ***Variable Insertion:***
(Det) Pred $\longrightarrow$ (Det) [Pred $\lambda y \,.\, y = x$]

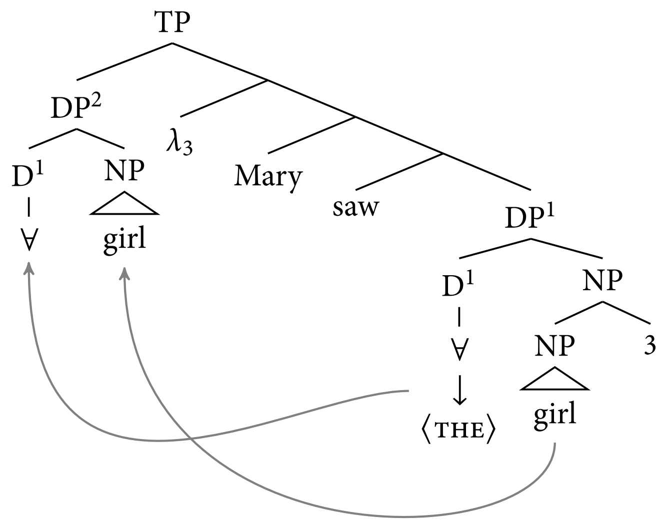
b. ***Determiner Replacement:***
(Det) [Pred $\lambda y \,.\, y = x$] $\longrightarrow$ the [Pred $\lambda y \,.\, y = x$]

(Fox 2002, (10): 67)

To begin to understand the kind of multidominance I will be proposing in this paper, one can begin by observing that, after Trace Conversion, the contents of DP‡ are different from the contents of DP†, despite the fact that they are to be understood as identical to each other, by virtue of being copies of each other. Specifically, the index 3 is not present in DP‡. So, if one is being serious about not merging it into the structure countercyclically, the series of steps that must have taken place in this derivation is to: (i) first, build the entire DP† in the base position; (ii) then, take its D node and NP node separately and merge them to create a node DP‡, distinct from DP†; (iii) then merge it in the landing site.

My proposal is to implement this literally in the syntax, instead of saying that the whole DP† moves. One immediately conceivable way to do this is shown in (12). Here, D^{1} and NP are taken separately from inside the already built DP^{1} and merged together *again* to build DP^{2}. Arrows are used to show the direction in which Remerge takes place. I cosuperscript two things to indicate that, in the syntax, they are one and the same object in two different positions, that is, they have entered multiple dominance relations. Crucially, although D^{1} is shared between DP^{1} and DP^{2}, which makes them projections of the same head, they are distinct syntactic objects and not copies of each other. This is indicated by contrasuperscripting them.[7], [8]

(12)



2.2 Moving the entire DP

For the sake of completeness, I will address one point before proceeding. Given this architecture, one could ask: is there any circumstance under which an entire DP is simply moved from one position to another? My contention is that there is no principled reason for such movement to be impossible and, consistently with that, it does happen in cases of total syntactic reconstruction.

For instance, a sentence like (13) is usually described to have a high scope reading and a low scope reading of the indefinite. The high scope reading can arise from an LF that has *true* QR, that is, an LF that contains a movement structure with λ-binding, like the one in (12). It is true QR in the sense that the quantifier truly raises, as in, the effect of it being raised is semantically detectable. In (12), the movement is vacuous because *Mary* is not a quantifier; but in (13), the semantic effect of such a movement is detectable in that this reading can be paraphrased as there being a specific Austrian who is likely to win the gold medal. Crucially, however, another kind of movement — very different from the one in (12) — is possible, in

7. The question arises, of course, how the semantic content of D^1 is allowed differ inside DP^1 from inside DP^2. This is rigorously addressed by the structures in Fox & Johnson (2016), Johnson (2009, 2012, 2018), for whom the only thing that is shared between the higher and the lower DP is the NP and the higher DP is not internally merged, but externally merged. This is not immediately crucial for my purposes; so I will stick to my system. In this respect, my system is closer to the system in Fox (2017).

8. In this system, movement can still be triggered by Agree. QR is not the kind of movement where one talks about Agree and features, but if the movement shown in (12), instead of being QR, were some other kind of movement that is triggered by a probe looking for a D feature to merge with, then something needs to be said about how Agree takes place when DP^2 is merged into the structure. Here, both DP^1 and DP^2 being projections of the same head ensures that whatever featural needs could be satisfied by DP^1 can also be satisfied by DP^2. Therefore, there is no architectural hurdle against generating structures like this.

which the whole DP *an Austrian* is moved, as in (13b). When this happens, it is perfectly permissible for PF to delete the lower copy because that is what is consistently done at PF anyway, and LF is free to pretend the movement never took place by deleting the higher copy. This gives rise to the low scope reading of the indefinite, that in which the winner will likely be some Austrian or other.

(13) [An Austrian]$_1$ is likely to t_1 win the gold medal.
PF:
[an Austrian]$_1$ is likely to ~~[an Austrian]$_1$~~ win the gold medal

a. ✓$\exists$ > likely
LF:
[an Austrian] λ_1 is likely to [⟨the⟩ 1 Austrian] win the gold medal

b. ✓likely > $\exists$
LF:
~~[an Austrian]$_1$~~ is likely to [an Austrian]$_1$ win the gold medal

(Sauerland & Elbourne 2002, (1): 284)

In fact, in this system, the claim is precisely that the only possible way for the low scope reading to arise is through the LF in (13b). In the LF in (13a), it is impossible for the low scope reading to arise because the quantifier is not interpreted in its quantificational meaning in the lower position (true QR). The only way for that to happen is to there be no QR in the first place, which is the case in (13b). This insight is already present in Sauerland & Elbourne (2002), in that they emphasize that the movement of *an Austrian* happens at PF. All I am saying is to put it, among other things, in the narrow syntax and the attested linearization being relegated to PF.

What is impossible, however, is for the deletion pattern in (14) to be semantically interpretable. This is so for various reasons. Firstly, the predicate *win the gold medal* in the embedded clause does not get any external argument. That means both that the Theta Criterion will be violated and that semantic composition will crash at the next node up from *win the gold medal*. Secondly, because there is no λ-binding, the node to which the higher copy of *an Austrian* attaches is of type *t*, assuming *likely* does not take an external argument (which is the conventional wisdom about raising predicates). Therefore, even if such a deletion pattern arises at LF, the resulting structure will not be interpretable.

(14) ***Uncomposable LF:***
[an Austrian]$_1$ is likely to ~~[an Austrian]$_1$~~ win the gold medal

A question that arises at this point is what determines when a constituent can be LF-deleted the way *an Austrian* is LF-deleted in (13b). I would propose to think about this as what is the rule of semantic composition between two syntactic objects which have been merged in the narrow syntax. During the generation of the LF in (13b), the entire DP *an*

Austrian has been remerged with the root node. But that is not the only position where it finds itself in the structure, because it has a position downstairs as well. One way to think about how semantic composition proceeds in such cases is for the rule of semantic composition to be sensitive to the various syntactic positions of the objects it is combining: it is free to ignore either position, but at least one must be semantically composed, provided the higher position c-commands the lower position. I would formalize this as in (15).

(15) ***Rule for semantic composition:***
If α and β are two nodes in a tree, α dominates β, and α and β are sisters, then, during semantic composition, β can be interpreted in either α's sister position or in the position dominated by α, and must be interpreted in only one of these positions.

The effect of this rule is that, in (13b), when semantic composition is about to apply to *an Austrian* and the root node after movement has taken place, assuming α stands for *an Austrian*, and β stands for the root node, the root node is both sister to *an Austrian* and dominates the lower position of *an Austrian*. That is, the rule in (15) is applicable here. Now, following the rule, if the higher position is chosen for semantic composition, it leads to the uncomposable LF in (14), which then leads to crash. If the lower position is chosen, then semantic composition proceeds and the low-scope reading is derived.

Although this subsection is about moving entire DPs without separating their parts from each other, I will consider a theoretical possibility much akin to (12), which leads to pathological results but, in the end, leaves no effect on the predictions I make. I mention this here because understanding why such a pathological result is yielded and what are the consequences thereof will help shed light on the insight that the semantic composition rule in (15) seeks to capture. Notice that, during the construction of the LF in (13b), an alternate route could have been taken: instead of remerging the entire DP *an Austrian* into matrix SpecTP, just *an* could have been remerged separately with *Austrian*, creating a node distinct from the already formed node dominating *an Austrian* in embedded SpecTP, and this entirely new node could have been merged into matrix SpecTP. That is, something exactly like the derivation in (12) could have happened, but without the λ-insertion and the Predicate Abstraction. Again, there is nothing in principle that prevents this structure from being generated. However, once such a structure is generated, crucially, (15) can no longer apply to that LF the way it can to (13b). This is because, in that LF, the node being merged with matrix $\overline{\mathrm{T}}$ is not also dominated by $\overline{\mathrm{T}}$, which is a necessary condition for (15) to apply to any structure. That is, if such a structure is generated, then the only interpretive option is to semantically interpret everything everywhere and to not delete anything — just as in (12). That leads to an LF just like (13b), except that the higher DP *an Austrian* must also be semantically composed with the rest of the sentence. And since, as explained for the case of (14), matrix $\overline{\mathrm{T}}$ is of type *t*, interpretation will crash because of type reasons. That is, such structures are eliminated by the pathological semantics they necessarily lead to, and nothing else needs to said about them.

2.3 Reconstruction in *wh*-questions

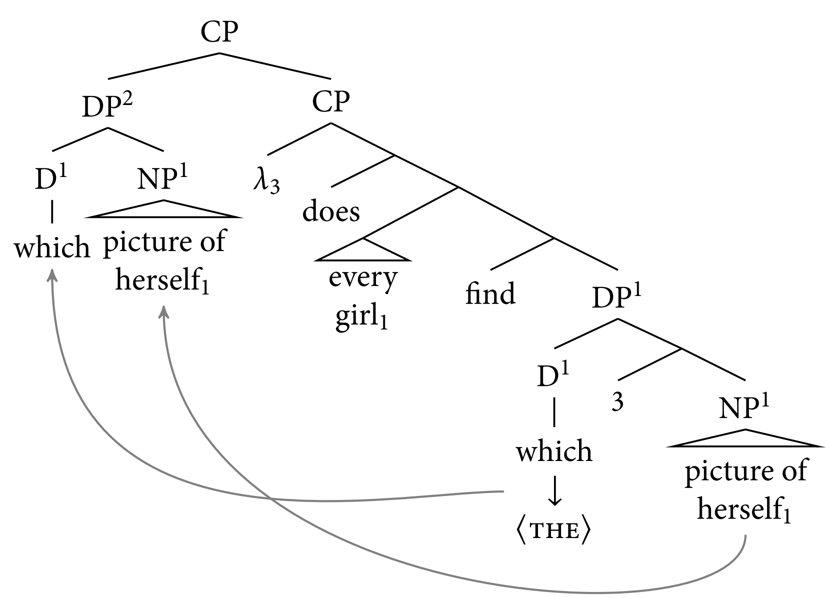
To understand how this way of envisaging DP movement makes a difference for the understanding of reconstruction in *wh*-questions, there are some basic assumptions that need to be made related to the semantics of *wh*-questions. In this paper, as far as the semantics of *wh*-questions is concerned, I am assuming the framework in Heim (2019). The part of her system that is immediately relevant for my purposes is that *which* is analyzed as a generalized existential determiner. The rest of her semantics is strictly concerned with question semantics (regarding issues known since Engdahl 1980, 1986), which will not be relevant for my purposes. So, setting aside the details specifically related to question semantics, the structures I will consider for *wh*-movement will be, broadly speaking, parallel to (12). That is, instead of a quantifier headed by *every*, the moving element will be a *wh*-phrase headed by *which*, and the meaning of *which NP*, where NP stands for the restrictor of *which*, will be that of a generalized existential quantifier. I will assume the usual lexical entry for this quantification determiner, given in (16).

(16) $[\![\text{which}]\!] = \lambda P_{et} \,.\, Q_{et} \,.\, \exists x[P(x) = Q(y) = 1]$

That said, for the sentence in (1a), repeated in (17a), (17b) should be the appropriate LF. (17) is entirely parallel to (12). Just as in (12), the quantifier *which* and its NP restrictor *picture of herself* merge with each other to create DP^2, after being part of the same DP^1. DP^1 and DP^2 are distinct syntactic objects, although they are projections of the same syntactic head *which*.

(17) a. Which picture of herself$_1$ did every girl$_1$ find?

b.



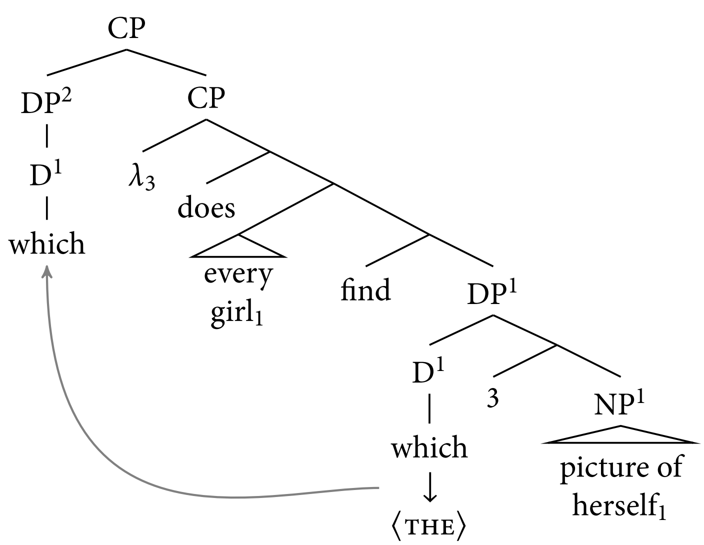
However, this was not the goal. Recall that, for (1a), repeated below, the target LF I am considering is like (1b), where *picture of herself* is interpreted only in the lower copy, so that *herself* can be bound by *every girl*. And my main motivation behind merging elements of a moving DP with each other to create a new node is to highlight the possibility of the higher and lower copy in DP movement to look different from each other, as in (1b). Furthermore, (15) cannot come to rescue here to delete *picture of herself* inside DP^2 because *picture of herself* and the lower CP projection are not sisters.

(1) a. Which picture of herself_1 did every girl_1 find?

b. [which] did every girl_1 find [picture of herself_1]

But if (1b) is taken at face value, then it would seem that only *which* moves out of the lower DP, and then, instead of being merged with any NP a second time, it is directly merged to the CP, as in (18). This does look a lot like (1b). But here, *which* is unrestricted, that is, it is missing its first argument (among its two $\langle e, t\rangle$-type arguments; see (16)).

(18)



In fact, this is the route Heim (2019) takes. Heim proposes a unary, unrestricted semantics for *which*, as in (19), so that structures like (18) can compose.[9] There are good reasons for giving such a semantics for *which*, one of them especially being the reconstruction fact in (3a): that the NP restrictor of a moving *wh*-phrase is interpreted in its lower copy, which is why there is a disjoint reference effect between *Alma* and *she* in *Which aspect of Alma does she despise?*.

(19) ***Unary* which**
$[\![\text{which}]\!] = \lambda P_{et} \,.\, \exists x_e [P(x)]$

9. Her reasons relate specifically to the semantics of functional *wh*-questions. The functional reading of (1a) becomes apparent when it is answered by saying something like *her wedding picture*.

But it is not always the case that the NP restrictor of the *wh*-phrase is semantically interpreted only in its lower position, for instance, in (20). (20a) is a case of long-distance *wh*-question, and (20b) is a case of embedded *wh*-movement. In both of them, the moving *wh*-phrase contains an anaphor that is bound by an R-expression in the highest clause in the sentence, that is, outside the clause in which the *wh*-phrase is base-generated. The sentences are nevertheless acceptable with the coindexation shown. That should not have been the case if *picture of herself* did not get "close enough" to *Mary* during the course of *wh*-movement.[10] That means that, unlike in (1a), the NP restrictors of the moving *wh*-phrases in (20) are not *in situ*. Therefore, a consistently unary semantics for *which* is not possible.

(20) a. Which picture of herself$_1$ with her lover did Mary$_1$ realize that her husband$_2$ had found out?

b. Mary$_1$ couldn't figure out which picture of herself$_1$ with her lover her husband$_2$ had found out?

One could take the messier route of positing both the unary and the binary lexical entry for *which*. But I will not choose that option here. What seems to be a cleaner view of *which* is that, like every quantifier, it is consistently binary and always needs a restrictor.

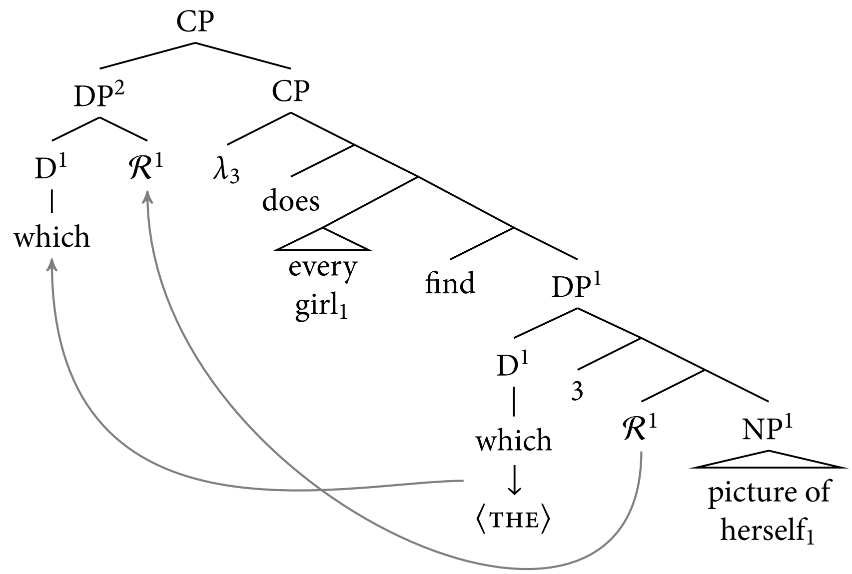
That means that something needs to be said about (18). My proposal is to exploit something that have been ignored in the structures so far, but is independently necessary for natural language quantifiers anyway: domain restriction (*e.g.*, von Fintel 1994, among others). By domain restriction, I will mean a contextually supplied predicate that needs to be intersected with restrictors of quantifiers in order for the perceived meaning to arise. For instance, *Every student passed* hardly ever means that every student that exists passed; instead, it means something weaker: when there is a contextually salient set of student, *e.g.*, a class that is salient in the conversation, the sentence can be taken to mean that every student in that class passed. That means that a predicate meaning "λx_e . x is in the contextually salient class" is being intersected with the meaning of *student*. The same is true for *wh*-questions: *Which students passed?* does not require the addressee to answer with a list of every existing student that passed.

My strategy, then, is to have a structure like (21), where $\mathcal{R}$ stands for a domain restriction variable. For my purposes, it does not matter exactly what the denotation of $\mathcal{R}$ is, because its point is to vary from context to context. The movements shown are again completely parallel to the ones (12): the quantifier *which* and the domain restriction predicate $\mathcal{R}$ are targeted separately by Merge, they are merged together to build the new node DP^2, which is merged into SpecCP. This is, finally, the kind of LF that was necessary: one in which *which* takes a restrictor and *picture of herself* is *in situ*, thereby making binding possible. This is my final proposed LF for (1a). I will call such structures $\mathcal{R}$*-sharing structures* because the $\mathcal{R}$

10. The exact definition of "close enough" has to do with what a binding domain consists in. For my purposes here, I will assume the most blunt version of the notion: that every clause is a binding domain, and once something escapes that clause, it is in the next binding domain higher up.

node is shared between a higher and a lower DP. In other words, I propose to model what is conventionally called *reconstruction* in *wh*-movement as $\mathcal{R}$-sharing.

(21)

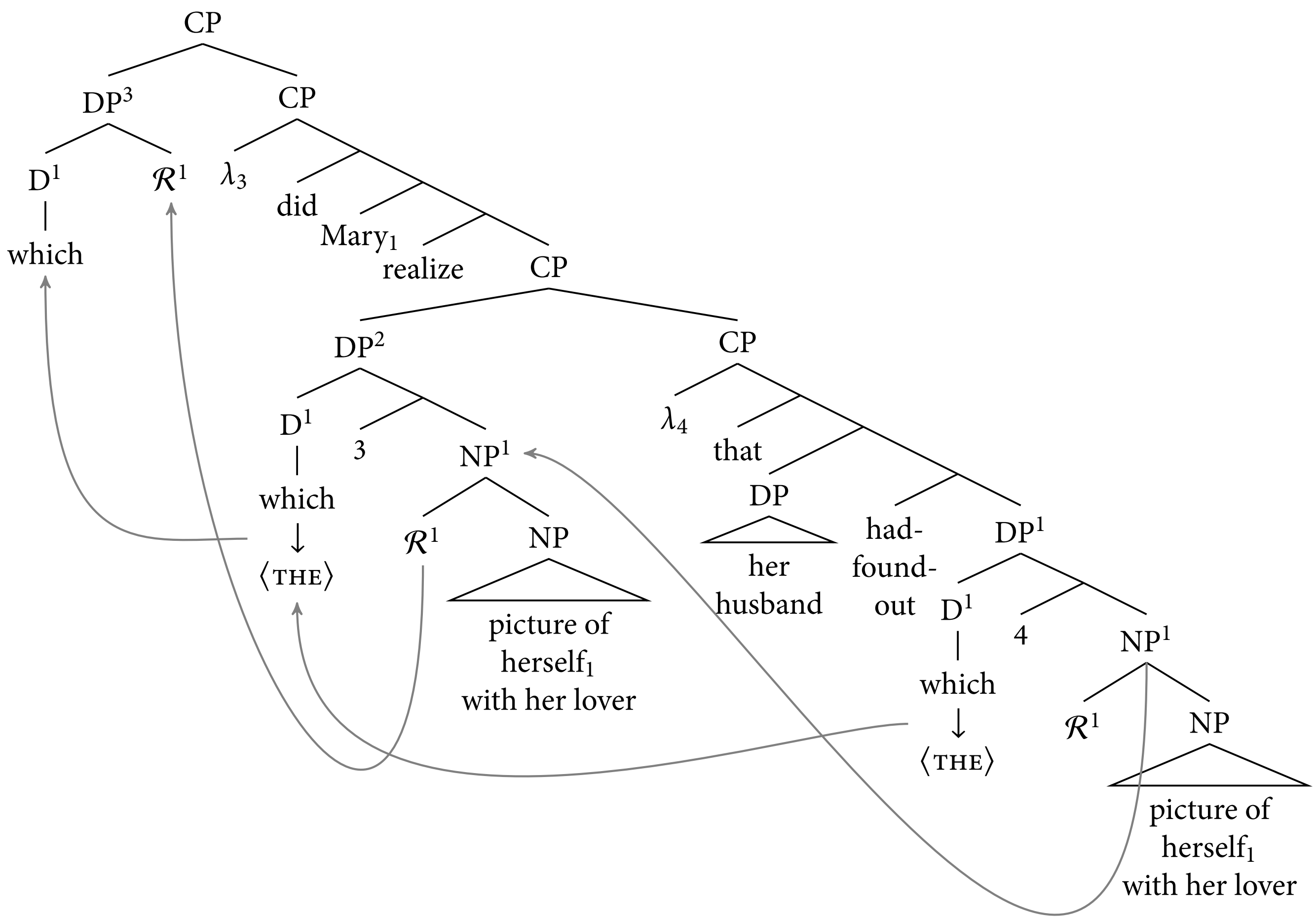


For the sake of completeness, I will address one more thing before proceeding. $\mathcal{R}$-sharing structures are not the only possible structure for *wh*-movement. A structure like (17b) is also possible, as long as the NP restrictor of the *wh*-phrase does not need to be "reconstructed". For instance, in *Which book did John read?*, the exactly parallel structure to (17b) — where *book* is shared between the higher and the lower DP — is completely feasible and causes no interpretive problem at LF.

In fact, pertinently, there are cases where at least one step of *wh*-movement must not have $\mathcal{R}$-sharing. For instance, in the cases in (20). Exactly as one would say under a mainstream notion of movement, it can be said, under the framework I am proposing, that there is successive cyclic *wh*-movement of *which picture of herself with her lover* from inside the embedded clause to embedded SpecCP, then to matrix SpecCP. During the first movement, technically, two different derivations are possible. If this movement involves $\mathcal{R}$-sharing as in (21), then *picture of herself with her lover* will stay *in situ*, and will not have moved close enough to *Mary* for *herself* to get bound by it, thereby violating Condition A. So, such a structure will be generable but not interpretable. But if this movement does not involve $\mathcal{R}$-sharing, instead, involves sharing of the entire NP restrictor *picture of herself with her lover*, as shown in (22), then the NP does get close enough to *Mary* for Condition A to be satisfied. However, in the next step of movement, there must be $\mathcal{R}$-sharing as in (21) because otherwise, the same problem that arises in (17b) arises again. Therefore, for the same reason, $\mathcal{R}$-sharing is necessary in the second step.[11]

11. Although $\mathcal{R}$-sharing in the first movement would lead to Condition A being violated in (22), that will not be the case in every such case of successive cyclic movement. For instance, both of the sentences in (i) are completely parallel to the sentences in (20), except that the anaphor is of a suitable gender to corefer with the subject of the embedded clause. These cases are reportedly ambiguous between the anaphor being coreferent

(22) ***LF for (20a):***



2.4 Linearization

So far, I have proposed a system in which, for (1a), the structure in (21) is generated directly in the narrow syntax. The benefit of this is that LF gets to see a structure that is ready-made for interpretation and no deletion operations take place. But the obvious problem this gives rise to is how to derive the attested linear order from (21) at PF because, given all the c-command relations between the nodes in the structure, there is no mainstream understanding of linearization (in the vain of Kayne 1994, for instance) that produces the linear order in (1a) from the structure in (21).

with the subject of the matrix clause and the subject of the embedded clause. My system predicts this. Here, the kind of $\mathcal{R}$-sharing structure for the first movement that gives rise to Condition A violation for (22) still allows *himself* to be bound by *his husband*.

(i) a. Which picture of himself$_{1/2}$ with his lover did John$_1$ realize that his husband$_2$ had found out?

b. John$_1$ couldn't figure out which picture of himself$_{1/2}$ with his lover his husband$_2$ had found out.

My proposal is to exploit the multiple dominance relations that the shared $\mathcal{R}$ node enters into. Notice that, in (21), *which* and $\mathcal{R}^1$ are sisters, $\mathcal{R}^1$ and NP^1 are sisters, but *which* and the NP^1 are not sisters. That is, under my system, the sisterhood relation is not closed under transitivity. The result of this is that, in every structure I have shown, even if *which* in the final landing site does not have a phonologically contentful NP restrictor (that is, not $\mathcal{R}$, which is not phonologically contentful), the $\mathcal{R}$ node inside that DP is always sister to a contentful NP restrictor. The net result of my algorithm will be to find this contentful NP and pronounce it immediately to the right of the pronounced determiner *which*.

To get this result, I propose the linearization algorithm in (23), which I will call the *Utterable Sister Algorithm (USA)*. The name is supposed to reflect the relevance of whether the NP sister of a determiner is phonologically contentful. I take the instruction in (23a) to stand for the basic fact about of English that there must be an overt NP restrictor immediately to the right of a determiner. My goal does not include providing an explanation for this. So, I will simply state this.

(23) ***The Utterable Sister Algorithm (USA)***

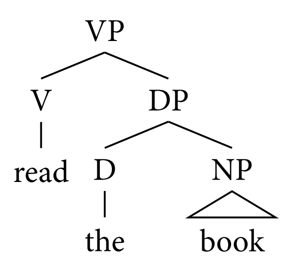
a. In English, every phonologically contentful determiner must have a phonologically contentful NP restrictor pronounced immediately to it s right.

b. If there is a phonologically contentful restrictor NP dominated by the mother of the phonologically contentful determiner, then that NP is to be pronounced immediately to the right of the determiner.

c. If there is not and instead, the phonologically contentful determiner c-commands an $\mathcal{R}$ node, then the phonologically contentful NP that is sister to this $\mathcal{R}$ node is to be pronounced immediately to the right of the phonologically contentful determiner.

The instruction in (23b) seeks to capture the fact that the DP in (24a) is pronounced as *the book*: there is a phonologically contentful NP restrictor dominated by the mother of this determiner, which is *book*. Therefore, this NP is pronounced immediately to the right of *the*. The entire linearization is given in (24b). I use the symbol "<" to mean "linearly precedes".

(24) a.



b. ***Linearization of VP***
read < the < book

The most crucial instruction is the one in (23c). This specifically addresses the question of how to go about pronouncing a DP which does not have any phonologically contentful NP restrictor, for instance, DPs like DP^2 in (21). Recall that the possibility multidominance removes transitive closure from the notion of sisterhood. In (21), this results in the following: *which* is sister to $\mathcal{R}^1$, $\mathcal{R}^1$ is sister to NP^1 *picture of herself*, but *which* is not sister to NP^1 *picture of herself*. (23c) is specifically intended to look past this absence of sisterhood relation between *which* and NP^1, through the connecting link of $\mathcal{R}^1$ between DP^1 and DP^2.

Let us break down how the reasoning in (23c) proceeds. It is ensured by (23a) that every phonologically contentful determiner in English must have a phonologically contentful NP restrictor immediately to its right. In (21), DP^2 does not contain a phonologically contentful NP restrictor because *picture of herself* is *in situ*. Therefore, the c-command relations between the various nodes of the structure in (21) does not place *picture of herself* immediately to the right of *which*, that is, to the left of the string *did every girl find*. This means that (23b) is not enough for successfully linearizing DP^2. So, the algorithm moves on to (23c). According to this instruction, *which* is sister to $\mathcal{R}$, therefore, $\mathcal{R}$'s phonologically contentful NP sister *picture of herself* — wherever it is in the tree — must be pronounced immediately to the right of *which*. This yields the attested linear order *which* < *picture* < *of* < *herself* < *did* < *every* < *girl* < *find*. That is, this algorithm makes it possible to generate structures with *in situ* NP restrictors, and still pronounce the NP restrictor in the left-peripheral position.

At this point, it would be reasonable to point out that this is way too stipulative, in that (23c) says exactly what needs to be said a bit too perfectly. This might be a legitimate objection, but only as long as one restricts attention to simple cases of reconstruction in *wh*-movement, as in (21). Once this algorithm is applied to more complex cases, it starts showing a certain systematicity and explanatory power. I turn to this now. The discussion that follows contains the empirical justification for the entire framework I am proposing here — both $\mathcal{R}$-sharing and the USA. This motivation involves data that are traditionally discussed in terms of LM, as I have explained in the introduction, in the context of (3b). So, before going to the motivating data themselves, I will first show how cases like (3b) can be captured under the system I am advocating here, and then show how the motivating data are a natural prediction of the combination of the syntax and the USA.

2.5 Interim summary

To take stock, I model cases of so-called *reconstruction* in *wh*-movement as the NP restrictor of the moving *wh*-phrase being left *in situ*, and instead, the domain restriction variable $\mathcal{R}$ being shared between the landing site and the base-generation site. This fulfills one of the promises I made in the introduction: that of modeling reconstruction by keeping NPs *in situ* already in the narrow syntax. The problem that this raises is that of pronunciation. That is, if (1a) must have a structure like (21) throughout narrow syntax and the interfaces, then how is it pronounced as if the entire *wh*-phrase, along with its NP restrictor, has undergone movement? As I have said before, this can be solved by writing a complex but systematic linearization algorithm. This is what I have called the *Utterable Sister Algorithm (USA)*, whose job is basically to pull the NP restrictor of the *wh*-phrase to the left of everything.

Since such a view — especially one that involves such an apparently odd linearization algorithm — might seem stipulative. So, before getting to the empirical motivation for this approach, I will say a few words on why this is at least conceptually superior to moving an entire *wh*-phrase. My main point consists in the proposal that there is no deletion operation at LF, unless not deleting results in a semantically uncomposable structure. This is already conceptually superior to any approach that proposes scattered deletion at LF, as I have outlined in the introduction. To repeat, it is understandable why deletion happens at PF: otherwise, every copy of everything would have to be pronounced, leading to far too unwieldy strings. But, there is no such consideration for LF. That is, there is no principled reason for which parts of a structure should be deletable. For instance, consider the deletion operations in (2c), where *picture of herself* is deleted in the higher copy and *which* is deleted in the lower copy at LF. But one could ask: why is *picture of herself* being deleted? It is of the write type, it can therefore combine with *which*. In fact, if it is deleted, *which* remains unrestricted and that should make the structure semantically uncomposable. This creates a distinct impression that LF deletion is really a description of facts, not an analysis. Under my system, there is no LF deletion, except when such deletion is triggered directly by the rules of semantic composition, *e.g.,* (15). For instance, in (13b), there is no way for *an Austrian* to compose with its sister because *is likely* takes only a clausal argument, which means that, after combining its clausal argument, it is not of a type that a generalized quantifier can combine with, without λ-abstraction. But This is why, I believe, the system I am proposing makes more conceptual sense.

3 Further motivation for the proposal: stacked modifiers

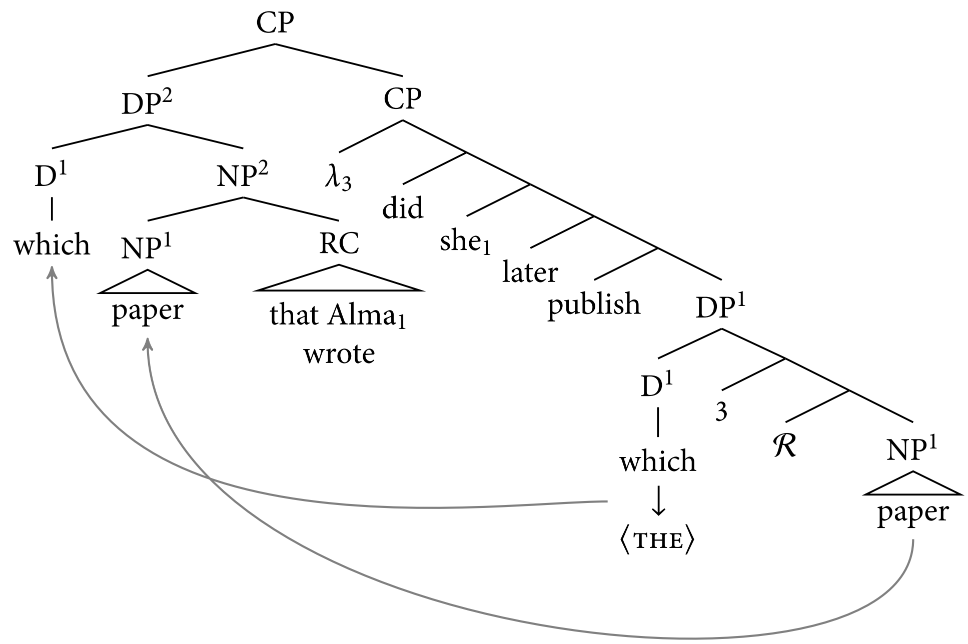
3.1 Background on antireconstruction

So much for reconstruction in *wh*-questions. Now, for the sake of completeness and to make the following discussion easier, it will be helpful to see how antireconstruction in *wh*-questions is modeled in this system. By *antireconstruction*, I mean the lack of a disjoint reference effect between *Alma* and *she* in (3b), repeated below. Here, the entire NP restrictor does not reconstruct, because coreference between *Alma* and *she* is possible. As I have said before, one traditional account of this effect is late merging the RC *that Alma wrote* to the NP restrictor of the higher copy of the *wh*-phrase (see (4)), and another is scattered deletion/Neglect (see (5)).

(3b) Which **paper** that Alma$_1$ wrote did she$_1$ later publish?

The goal of this paper is to model *wh*-movement in such a way that neither countercyclic Merge nor scattered deletion at PF is necessary. One half of this goal is to model reconstruction and the other half involves antireconstruction. Reconstruction, I have already addressed: they involve $\mathcal{R}$-sharing structures as in (21). Put differently, they involve structures in which there is more stuff in the lower DP than in the higher DP. Pursuing this insight, antireconstruction, then, would involve structures in which there is less stuff in the lower DP than the higher DP. For instance, (25). This is not my innovation at all. Structures like this have been elaborately developed in Fox & Johnson (2016), Johnson (2009, 2012, 2018). The point of having it here is to help the reader understand how reconstruction and antireconstruction are literally the converse of each other right from the narrow syntax and set the stage for the motivating data.

(25)



Here, D^1 and NP^1 are merged into the new node DP^2, but notice that they have not been merged with each other while building DP^2. First, NP^1 is merged with the RC *that Alma wrote*, to form the node NP^2. It is this node NP^2 that is merged with D^1 to form DP^2. That is, the RC is not present inside DP^1; it is only present inside DP^2. This explains why *Alma* and *she* can corefer without giving rise to a Condition C violation: because *she* never c-commands *Alma*. Notice that this is the mirror image of the kind of structure that is necessary to model reconstruction, as in (21). In (21), *picture of herself* must be present only inside DP^1, and not inside DP^2 because *herself* needs to be bound by *every girl*. In (25), the RC *that Alma wrote* must be present only inside DP^2 and not inside DP^1 because the lack of a disjoint reference effect needs to be captured.

Also notice that this, again, is conceptually superior to the scattered deletion approach in (5). There, *which paper that Alma wrote* moves as a unit to SpecCP, and then, at LF, *that Alma wrote* is deleted in the lower copy, but not deleted in the higher copy. In this case, under a scattered deletion approach, there is even less reason for forcing such a deletion operation to apply. This is because, if it does not apply, then nothing technically goes wrong; the only thing that happens is that, because of Condition C, coreference between *Alma* and *she* becomes impossible. But even in that case, the resulting interpretation with contraindexation between *Alma* and *she*, leads to a perfectly interpretable meaning: one in which *Alma* is not who *she* refers to. So, technically, the result should be a disjoint reference effect between *Alma* and *she*. But in a framework that has the option to put the RC only in the higher DP, as in (25), these considerations do not arise: interpretation is completely systematic, deletion never even come into question.

3.2 Stacked modifiers: LF and PF

Let us take stock. In the introduction, I presented the sentence *Which picture of herself did every girl find?*, repeated below, as requiring an LF like (1b), to ensure that *herself* can be bound by *every girl*. In the preceding section, I have proposed a way to model this through sharing $\mathcal{R}$ between the higher and the lower DP and thus allowing the NP restrictor *picture of herself* to stay *in situ*. This gives rise to the problem of linearizing *picture of herself* to the left of *did every girl find*. My solution to this problem of linearization was the USA. The USA basically forces *picture of herself* to be pronounced to the left of *did every girl find*, no matter where it is in the tree.

(1) a. Which picture of herself$_1$ did every girl$_1$ find?

b. [which] did every girl$_1$ find [picture of herself$_1$]

This way to go about the problem of reconstruction in *wh*-questions does seem quite stipulative. My goal in this section is to argue that this is not so stipulative after all, and, in fact, it comes with explanatory power. One part of my approach concerns generating the semantically necessary structures in the narrow syntax directly. This, in itself, may not seem all that stipulative, because, if Merge is free and various options can pan out in narrow syntax, then the structures I am proposing arise in the narrow syntax could indeed arise there. In fact, preventing them might seem stipulative. What does indeed seem more stipulative is the linearization algorithm, in that it is quite odd: it basically instructs PF to find a pronounceable NP restrictor, wherever it is, and, then, stop. Thus, it ends up doing exactly what it needs to do, a bit too perfectly, and does not seem to be explaining anything. What I will argue in this section is that this very strict linearization algorithm produces systematic results elsewhere, which are independently desirable.

To argue this, I will rely on (26). This is an example based on structurally parallel examples in Sauerland (1998; (2.37-2.38): 52; (2.40b-c): 53). In each of the sentences in (26a)-(26b), in (26), there is *wh*-movement of a DP. The NP restrictor of the DP has two modifiers. One of them hosts the R-expression *Mary* that is coreferent with the pronoun *she* in the question skeleton, and the other hosts a variable bound by the quantifier *every boy/professor* in the question skeleton. I've schematized the relevant parts of this in (27). Under an LM-based account, the pattern in (26) can be described as follows: when there are two adjuncts modifying an NP, only the outer RC can be LMed and, if it cannot — *e.g.,* because it contains a pronoun that must be bound by a quantifier downstairs — then the inner one cannot either.

The (c) sentences provide a baseline for the (b) sentences. The (c) sentences are different from the (b) sentences in that, the placements of the feminine pronoun and the R-expression have been swapped, which allows the inner modifier to not have to be late merged in order for coreference between the feminine pronoun and the R-expression to arise.[12]

(26) a. [Which [[assignment [submitted late by him$_2$]] that Mary$_1$ wasn't happy with]] did she$_1$ ask every boy$_2$ to resubmit?

b. *[Which [[assignment [submitted late by Mary$_1$]] that he$_2$ wasn't happy with]] did she$_1$ ask every professor$_2$ to reconsider?

c. [Which [[assignment [submitted late by her$_1$]] that he$_2$ wasn't happy with]] did Mary$_1$ ask every professor$_2$ to reconsider?

(27) [which [[NP MOD$_1$] MOD$_2$]]$_3$ ... t_3?

Therefore, if the pattern in (26) is to be derived under a mainstream approach of movement (by moving the whole *wh*-phrase and then applying scattered deletion at LF), there must be some constraint on scattered deletion that predicts only the following to be the permissible and impermissible deletion operations. But there is no principled reason for (28c) to be an impermissible LF, as opposed to the other two. That is, (28) only describes (26); it does not explain it. This, therefore, is one datapoint that a scattered deletion approach fails to capture, unlike in the case of simpler cases like (1a) and (3b).

(28) ***PF (same for all LFs):***
[which [[NP MOD$_1$] MOD$_2$]]$_3$
... ~~[which [[NP MOD$_1$] MOD$_2$]]$_3$~~

12. Sauerland (1998) explains the ungrammaticality of the only-outer sentences by modifying the original Lebeuxian LM explanation a little bit. Borrowing what Tada (1993: 63-70) says about similar examples in Japanese, Sauerland says that, although LM is countercyclic, there's a limit to its countercyclicity. The reason, Sauerland says, (i) is grammatical is because when LM happens to the SpecCP position, that instance of LM is cyclic with respect to that SpecCP. But, in the only-outer sentences, that can't be the case because the first RC must be merged in its base position so that the pronouns in it can get bound by the universal quantifiers, and then, because of this constraint, when LM is to happen in SpecCP, it must apply to a node rather deeply embedded inside this SpecCP, unlike in (i), and this countercyclicity with respect to the SpecCP where LM is happening is responsible for the ungrammaticality of the only-outer sentences.

(i) Which argument that John$_1$ published did he$_1$ later disprove?

However, there's some substantial vagueness in this explanation. That is, how is it that there's no countercyclicity in the LM involved in (i)? That is, this LM must happen to the NP embedded inside the DP sitting in the SpecCP, not to the DP itself. Then, how is the countercyclicity involved in the only-outer sentences also not involved in (i)? For these reasons, I will not consider this possibility as a viable analysis in this paper. Also see Fox (2017), who argues that, if LM must be allowed to be highly countercyclic, because there are cases where modifiers seem to be LMed inside islands.

a. ***LF (equivalent to LMing only the outer modifier; permissible):***
[which [[NP $\cancel{\textsc{mod}_1}$] $\textsc{mod}_2$]]$_3$
. . . [~~which~~ [[NP $\textsc{mod}_1$] $\cancel{\textsc{mod}_2}$]]$_3$

b. ***LF (equivalent to LMing both modifiers; permissible):***
[which [[NP $\textsc{mod}_1$] $\textsc{mod}_2$]]$_3$
. . . [~~which~~ [[NP $\cancel{\textsc{mod}_1}$] $\cancel{\textsc{mod}_2}$]]$_3$

c. ****LF (equivalent to LMing only the inner modifier; impermissible):***
[which [[NP $\textsc{mod}_1$] $\cancel{\textsc{mod}_2}$]]$_3$
. . . [~~which~~ [[NP $\cancel{\textsc{mod}_1}$] $\textsc{mod}_2$]]$_3$

The benefit of the system I am proposing here is that it can capture cases like (26) without any stipulation, and this happens in the following way. Just as in the case of (1a), where narrow syntax produces the structure in (7), repeated below, I will propose that, in order for the binding relations shown in (26b) to be established, narrow syntax itself must generate the LF like one in (29). In a configuration like this, the quantifier can bind the pronoun (just as *the sofa* can bind *herself* in (7)). The crucial distinction between (7) and (29) is that in the latter, as opposed to the former, $\mathcal{R}$ is never merged with *which*; downstairs, it is merged with the NP, and upstairs, it is merged with the modifier containing the R-expression, which has no occurrence in the c-command domain of pro_1.

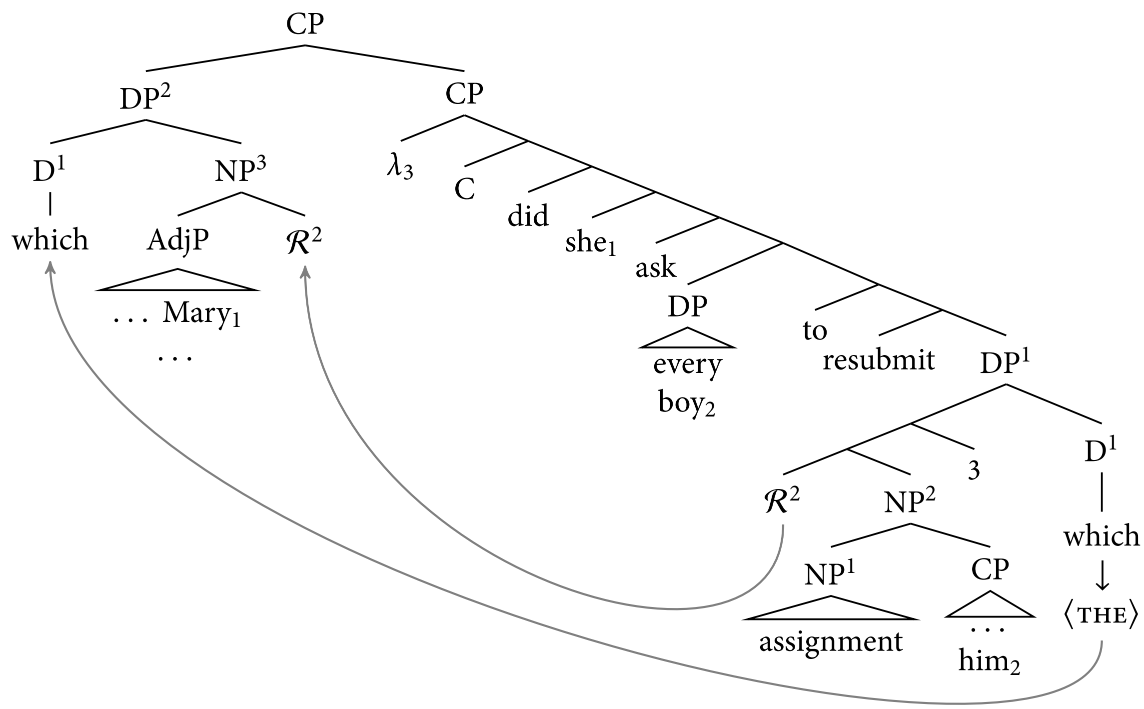
(7) [which $\mathcal{R}^1$] did every girl$_1$ find [THE $\mathcal{R}^1$ picture of herself$_1$]

(29) ***LF:***
[which [$\mathcal{R}^1$ $\textsc{mod}_{R\text{-}expr_1}$]] . . . pro_1 . . . quantifier$_2$ . . . [THE [$\mathcal{R}^1$ NP $\textsc{mod}_{pro_2}$]]
where:
$\textsc{mod}_{pro_2}$ = NP containing a variable with the index 2 bound by the universal quantifier inside the question skeleton;
$\textsc{mod}_{R\text{-}expr_1}$ = NP containing an R-expression coreferent with the pronoun indexed 1 inside the question skeleton;
the numbers stand for indices.

Once such structures are generable, the outline of my account is the following. Each of the sentences in (26) involves a pronoun in one of the modifiers getting bound by a quantifier in the question skeleton. In order for the binding to take place, that part of the *wh*-phrase must reconstruct, that is, be present only in the lower DP, while $\mathcal{R}$-sharing between the higher and the lower DP takes place, just like in (21). Plus, when there is antireconstruction, that is, the lack of a disjoint reference effect, as in (26a) and (26b), the modifier containing the R-expression must be present only in the higher DP, just like in (25). The combination of these two structural constraints conspire to make sure that the USA fails to yield (26b) with the intended bound variable reading. That is, (26b) is constructively eliminated.

Let us unpack that. I will begin with the grammatical case in (26a). Here, *him* is bound by *every boy*; therefore, *every boy* needs to c-command *him*. Moreover, *she* and *Mary* can corefer; therefore, *Mary* must not be c-commanded by *she*, or otherwise, Condition C would be violated. That means that, for the first part, the structure needs to look like the $\mathcal{R}$-sharing structure in (21), and for the second part, it needs to look like the antireconstruction structure in (25). A structure that features both these properties is given in (30) for (26a). This allows *him* inside DP^1 to be bound by *every boy*, without putting *Mary* in the c-command domain of *she*. This is the product of narrow syntax, which is then fed to the USA. For the purposes of linearization, the phonologically contentful determiner is *which*. Crucially, there is no phonologically contentful NP dominated by DP^2. Therefore, the phonologically contentful sister of the $\mathcal{R}$ node inside DP^1, that is, the entirety of NP^2 is to be pronounced immediately to the right of *which*. This ensures the attested linear order *which assignment submitted late by him that Mary wasn't happy with*. There is no other string that can be generated by the USA. For instance, *which assignment that Mary wasn't happy with submitted late by him* is not a possibility because in order for this to have been the string, the USA would have needed to refer to just the NP^1 bit, then the AdjP bit, and then back to the CP bit. There is no room for such haphazardness in the USA.

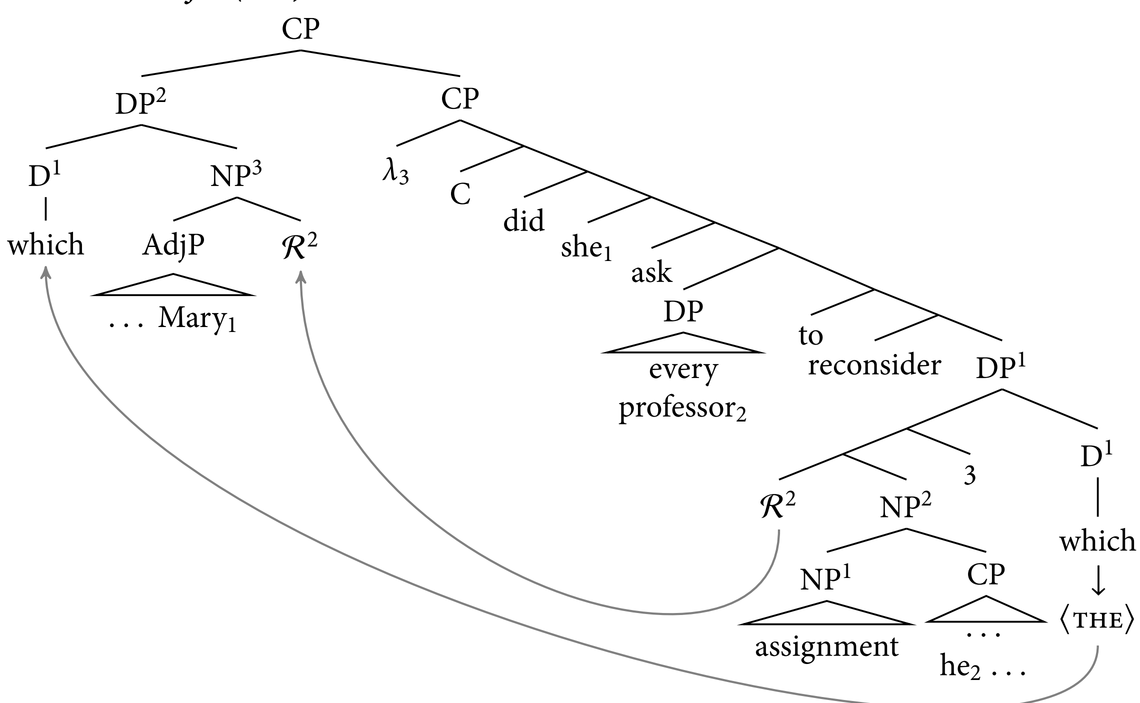
(30) a. ***Structure for (26a):***



b. ***Linearization:***
which < assignment < CP < AdjP < did < she < ask < every < boy < to < resubmit

The same kind of reasoning applies to (26b). Again, in this case, there is a modifier containing a variable bound by a quantifier in the question skeleton and another modifier containing an R-expression coreferent with a pronoun in the question skeleton. That is, the former needs to be present only in the lower DP and the former, in the higher DP. The only difference between (26a) and (26b) is that the linear order between the modifier with the bound variable and the one with the R-expression has been swapped. This is readily predicted under my account as follows. In order for the binding of *him* by *every professor* to take place and coreference between *she* and *Mary* to be possible, only the structure in (31) succeeds. All sorts of other structures, for instance, one in which the AdjP is only inside DP^1 and the RC is only inside DP^2 (let us call this alternative (31a′)) would have been generable — because nothing prevents them — but they would not be interpretable in the relevant way: that is, binding would not take place and there would be a disjoint reference effect between *she* and *Mary* because of Condition C. Now, the USA plays the next crucial part. When the USA applies to (31a), the only possible output is the one given in (31b), where the RC precedes the AdjP. This is because of the same reason as in (30a): because there is no phonologically contentful NP inside DP^2, the NP^2 node as a whole must be pronounced immediately to the right of *which*. In order for the linear order in (31b) to be the output, a structure like (31a′) needs to be generated; but that would not give rise to the relevant interpretation. Therefore, (26b) cannot be a grammatical sentence with the coindexations shown.

(31) a. ***Structure for (26b):***



b. ***Linearization:***
which < assignment < CP < AdjP < did < she < ask < every < professor < to < reconsider

This fulfills the promise I made in the introduction. I have given an account of reconstruction in *wh*-questions by generating the necessary structures directly in the narrow syntax and without any scattered deletion taking place at the interfaces. The linearization algorithm independently needed to linearize such structures turns out to account for additional puzzles, like the pattern with stacked modifiers discussed in this section. This indicates that, although the algorithm might seem stipulative in isolation, it is in fact explanatory when enough data are looked at.

4 Conclusion

It has long been known that *wh*-phrases can reconstruct for the purposes of binding. For instance, in order for the semantics of (1a), repeated below, to work, *picture of herself* must be present only in the lower copy; otherwise, *herself* will not be able to get bound by *every girl*. On the other hand, for the purposes of question semantics, *which* must be interpreted in the higher copy. That is, the interpretable LF for (1a) must look something like (1b).

(1) a. Which picture of herself$_1$ did every girl$_1$ find?

b. [which] did every girl$_1$ find [picture of herself$_1$]

This is traditionally achieved by moving the entire *wh*-phrase in the narrow syntax, and then applying scattered deletion operations at LF, to make sure that *which* is deleted downstairs and interpreted upstairs, and *picture of herself* is deleted upstairs and interpreted downstairs. My plea to the reader, in this paper, is to take the cue about syntax from semantics, rather than from pronunciation. That is, since we can tell that, for the sake of the right meaning to arise, the structure must be as in (1a), we should derive structures like this directly in the narrow syntax and not have any deletion operation anywhere. The trade-off for this approach is that a more complex linearization algorithm needs to be proposed, to make sure that *picture of herself* is pronounced immediately to the right of *which*, despite being present only in the lower copy. This is the first part of the paper. The next part of the paper consists of the empirical motivation for writing such a complex linearization algorithm. That is, my proposal is the following: yes, taking our cue from the semantics, instead of pronunciation, comes with the cost of requiring a complex linearization algorithm, but this algorithm, although complex, is systematic, strict, and leaves no room for any kind of stipulation, whereas a scattered deletion approach needs to be put under *ad hoc* constraints (see (28)). Therefore, if the proposal I am advocating in this paper is adopted, then, at the end of the day, we are at a better place, when it comes to giving a concrete formalization of what really goes on in reconstruction in *wh*-movement.

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